

A Spectroscopic Completeness Model for DESI dwarf galaxies

Background P1

One of the goals of my project is to compute the quenched fraction of low mass galaxies ($\text{LogMstar} < 9$) conditioned on environment.

Dwarf galaxies being intrinsically faint with large angular sizes have fainter fiber magnitudes resulting in more redshift failures than the typical BGS Bright cosmology sample. Furthermore, if filtered by some redshift success criterion (e.g., $\text{DChi2} > 40$), the spectroscopic sample will be biased towards star-forming thus biasing quenched fractions. Having a robust spectroscopic incompleteness model will be essential for measuring robust quenched fractions.

Additionally, the existing DESI redshift completeness model uses $\text{TSNR2} + r\text{-fiber mag}$, where TSNR2 is a metric for effective exposure time (does not depend on brightness or spectral type of source). There are also some assumptions of redshift distribution that go into the TSNR2 calculation that likely make it more suitable for the BGS cosmology sample. Our dwarf galaxy catalog mostly resides at lower redshifts ($z < 0.1$), with the smallest dwarfs (with the most interesting physics) at $z < 0.05$.

As we enter regimes of billions and billions of spectra, where we are no longer statistics limited but systematics limited, having a good understanding of spectroscopic completeness will be important.

Background P2

This motivates the exploration of a redshift completeness model specifically for our dwarf galaxy sample (or a model that takes into account spectral type). The key assumption we make is that the redshift success rate, at a fixed exposure time, is primarily influenced by the fiber magnitude, DCh₂, and the strength of spectra features. In case of no nebular emission, the fiber magnitude should correlate with SNR of stellar continuum and thus spectral feature strength. We were initially using H α EW/flux as a proxy for strength of spectral features as it is the most prominent emission line.

However, to make this framework useful for a wide range of galaxies and science cases, we explore a different way using NMF to characterize spectral features.

NMF = non-negative matrix factorization is in spirit similar to PCA, but differs in many important ways. The most relevant and important difference being, each component/template in NMF is non-negative (and thus coefficients too). Thus, every spectra is a non-negative linear combination of non-negative templates. This is more physical (as flux > 0) than PCA approach where templates and coefficients are non-negative and unphysical.

Q: Why not use a definition of redshift success rate (e.g., $D_{\text{chi}2} > 40$) and use the observed spec-z catalog to model the success rate as a function of parameters we care about?

A: This does not account for the incorrect redshifts entering, for example, into our $z_{\text{RR}} < 0.15$ sample that will bias the redshift success rate. This also does not correct for the redshifts that are not in our $z_{\text{RR}} < 0.15$ sample but have been kicked out ($z_{\text{RR}} > 0.15$) due to incorrect redshifts.

Below is a visualization of the above story. If not accounting for these, our redshift success rates will not be accurate ...

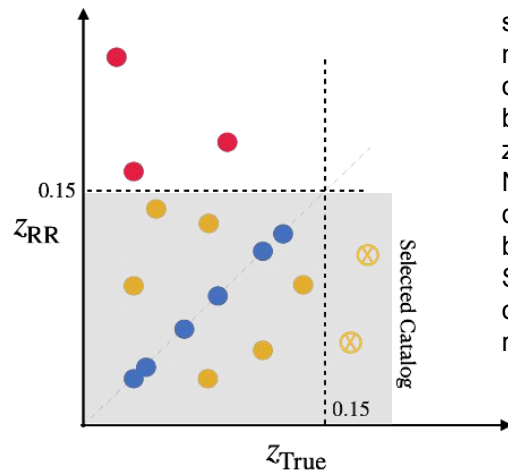
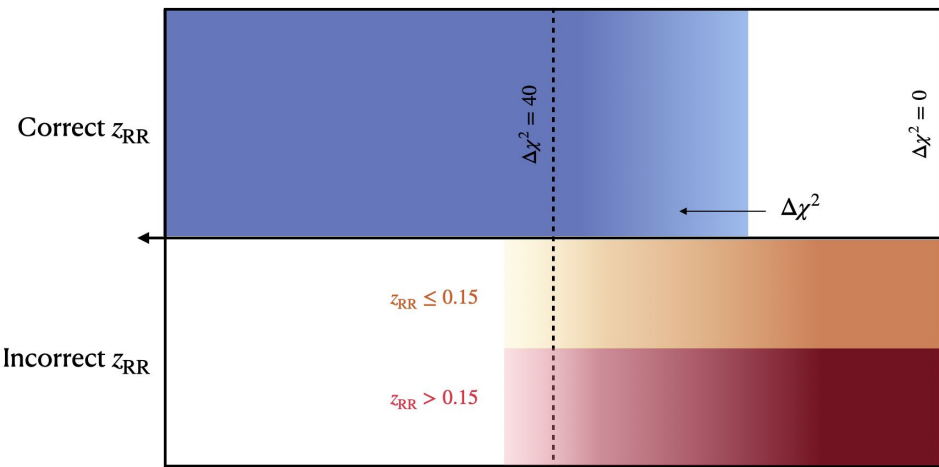


Fig: Imagine these are galaxies with same spectra but different normalization, redshifts and observed multiple times (this is our bin). This is showing the possible z_{RR} outcomes of such galaxies. Note the orange+red points here, due to incorrect z_{RR} , will leave “this bin” enter some other “bin”. Something similar happens the other direction too with incorrect redshifts entering into “this bin”

“Two roads diverged in a wood ...”

I think there are 2 ways to do build this completeness model.

Method 1 is simpler, but we end up throwing out data and we need to have our remaining galaxies sample the full physical parameter space we care to apply corrective weights.

Method 2 is more difficult. Here we use the entire spectroscopic catalog (below some $z_{RR} < 0.15$), however, we need to have a good understanding of the purity and completeness to appropriately weight each data point. The trickiest part of this method is that it relies on having a good understanding of the entire targeted sample, even galaxies at $z > 0.15$ that we do not care about. This is because we would need to know how frequently these galaxies are incorrectly entering into our bins.

I will explain in more detail both these methods. Both these methods rely on having a good spectral forward model. I believe I have quite a reasonable one right now.

Method 1

In this method, we only work with galaxies with confident redshifts (e.g., apply a $D_{\text{Chi2}} > 40$ cut).

We will model how frequently a galaxy with a given r-fiber mag and spectral type goes below $D_{\text{Chi2}} < 40$. That will be our redshift failure rate and then we can use that to weigh the data in our catalog. One would expect quenched, small EW galaxies to go below $D_{\text{Chi2}} < 40$ more frequently, in contrast to star-forming, high EW galaxies.

The key assumption here is that galaxies in our confident redshift catalog are sampling the entire quenched-SF parameter space. If, lets say, the D_{Chi2} cut is selectively cutting all quenched galaxies in some parameter space, we will not have any of those galaxies in our confident-redshift catalog to apply weights to. Or if some parameter space has a very high failure rate ($> \sim 90\%$), it is likely we do not have any of those galaxies in our catalog due to low statistics, again not allowing us to correct for it.

Note that in parameter space where failure rate is 100%, **both method 1 and method 2 cannot correct for that.**

This failure rate would be measured using simulations.

Method 2 (the road less travelled?)

In this method, we work with the entire spectroscopic catalog at $z_{\text{RR}} < 0.15$ (the redshift range in which our dwarfs exist). There will be many wrong redshifts in our sample, and we will be statistically taking care of that by appropriate up- and down-weighting. There are two weights that will be relevant here: i) purity weight and ii) completeness weight (or success rate).

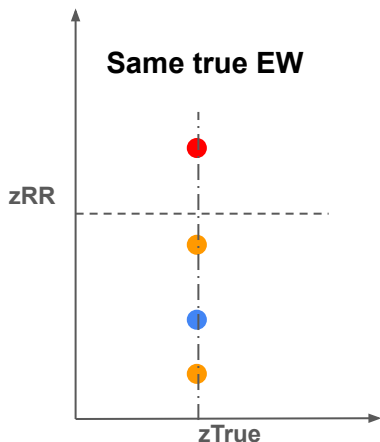
Purity weight is the probability of a galaxy in our catalog having an incorrect redshift. This is used for down-weighting and is measured as a function of the r-fiber mag, DChi2, and H α EW/flux (or spectral metric $\mathcal{S}$)

Completeness weight is the probability of a galaxy's redshift being **correct** given r-fiber mag and H α EW/flux (or spectral metric $\mathcal{S}$). This is used for up-weighting galaxies to correct for missing galaxies due to incorrect redshifts

In the next few slides, I will discuss a concrete that will provide more intuition on this.

Let us consider a toy example to get a better intuition on these weights. Imagine that we have 4 galaxies with the same **true** Halpha EW/spectra, redshift and r-fiber mag.

Out of these 4, 1 has correct redshifts and 3 have incorrect redshifts. 1 of these 3 had an incorrect redshift that placed it at $z \sim 1$ (so outside our $z_{RR} < 0.15$ cuts).

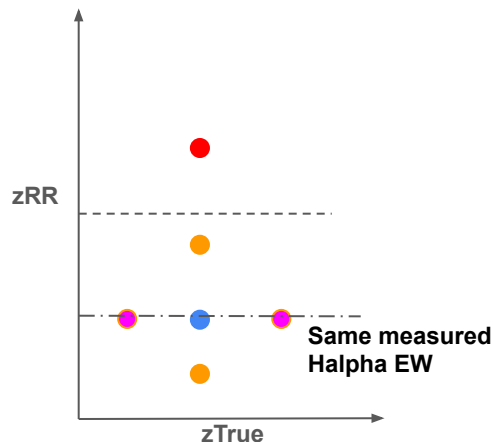


The 3 incorrect redshifts would thus have an incorrect measured Halpha EW. The 2 incorrect at $z_{RR} < 0.15$ would remain in our science analysis, but enter some other incorrect Halpha EW bin. The 1 incorrect at $z_{RR} > 0.15$ would not enter our science analysis at all.

How do we deal with the fact that 3 galaxies have been knocked out of our current Halpha EW bin due to incorrect redshifts? We use the completeness weight! Using simulations, we study how frequently there is a correct measured redshift for a galaxy with a given r-fiber mag and spectral type. In this case $\frac{1}{4}$.

The purity fraction would take care of the 2 incorrect redshifts in the other Halpha EW bin by statistically downweighting. More on this in next slide.

Note that just how we can have galaxies leave our Halpha EW bin due to **incorrect redshifts**, we can have galaxies with incorrect redshifts enter of Halpha EW bin too. In our toy example here, we have **2 galaxies** with incorrect zRR that enter our Halpha EW bin.



By modeling the purity weight as a function of r-fiber mag, DChi2 and metric for spectral strength, we will count these **3 galaxies** as effectively only 1 galaxy by appropriately downweighting each galaxy. The weights on these 3 galaxies need not be equal ($\neq \frac{1}{3}$) and can be different, but on average be $\frac{1}{3}$.

So how do we correct for these effects in practice? How do we go from 3 galaxies in our measured bin to 4 that are supposed to be in our bin?

It would be like this:

$$N_{\text{meas}} = 3 \rightarrow w_{p1} \cdot 1 + w_{p2} \cdot 1 + w_{p3} \cdot 1 = 1$$

Down-weighting each galaxy in our measured bin by the purity weights.

Let us suppose
these are the purity
weights

1/12

1/6

1/12

$$= 1 \rightarrow \frac{w_{p1}}{w_{c1}} \cdot 1 + \frac{w_{p2}}{w_{c2}} \cdot 1 + \frac{w_{p3}}{w_{c3}} \cdot 1 = 4 = N_{\text{eff}}$$

After correcting for impurities, we then correct for the lost galaxies by up-weighting by inverse of completeness weight (success rate)

Our
completeness
weight is 1/4 as
shown in prior
slide

1/4

1/4

1/4

More generally, we can write this as

$$N_{\text{eff}}(\text{some bin}) = \sum_i^{N_{\text{meas}}} \frac{w_{pi}(r_{\text{fib},i}, \Delta\chi_i^2, \mathcal{S}_i)}{w_{ci}(r_{\text{fib},i}, \mathcal{S}_i)}$$

$$N_{\text{eff}}(\text{some bin}) = \sum_i^{N_{\text{meas}}} \frac{w_{pi}(r_{\text{fib},i}, \Delta\chi_i^2, \mathcal{S}_i)}{w_{ci}(r_{\text{fib},i}, \mathcal{S}_i)}$$

There is a small subtlety here that I want to point out which may not be immediately obvious. In our observed catalog, we will never have access to the true H α EW/flux (or spectral metric $\mathcal{S}$). So in the above formula, $\mathcal{S}_i$ is actually $\mathcal{S}_{\text{meas},i}$ even when the purity and completeness weights are measured on $\mathcal{S}_{\text{true},i}$ as based on simulations.

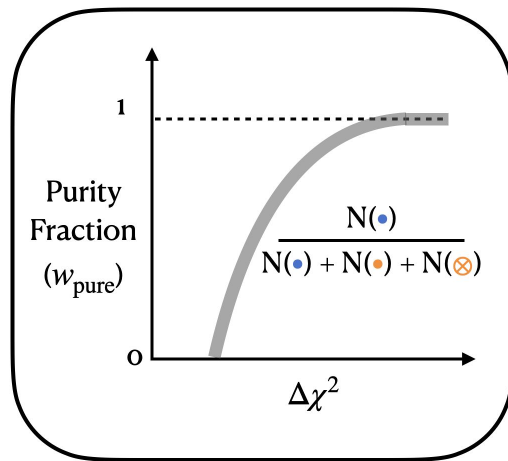
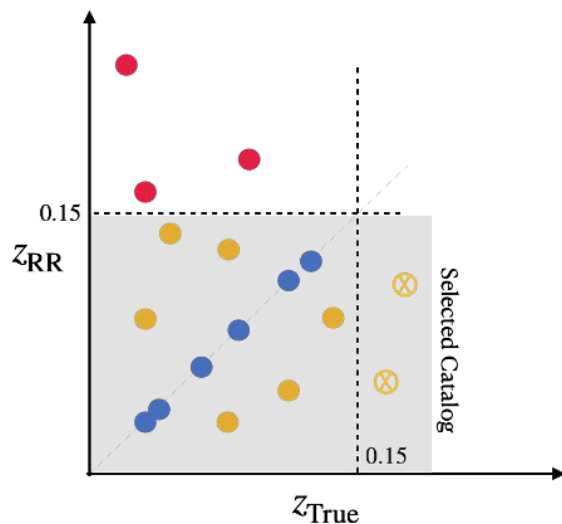
The idea here is that at some $r_{\text{fib},i}$, where we are getting predominantly noise, it does not matter what H α EW/flux or $\mathcal{S}$ we are measuring. It will be a meaningless quantity. In such scenarios, the

However, we could train our simulation to run on recovered galaxies too?

Two quantities relevant to Method 2 ...

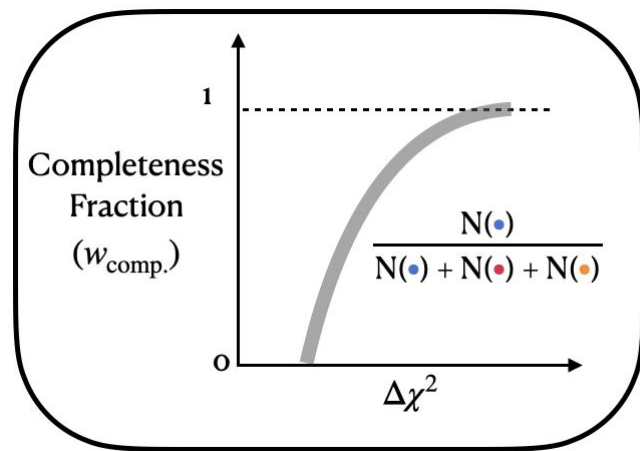
The purity fraction is defined as the fraction of galaxies **in our observed sample** that have a correct redrock redshift. This is specifically defined for galaxies with $z_{RR} < 0.15$.

Assume below are galaxies with same H α EW and r-fiber



The completeness fraction is defined as the proportion of low-redshift galaxies with correct redshifts. The inverse of this will be used to upweight galaxies to *correct for lost true low redshift galaxies*.

In other words, what we are trying to get at is for every correct z_{RR} low-redshift galaxy, how many incorrect z_{RR} low-redshift galaxies are we missing?



Metric for strength of emission lines

Till now, I have considered H α EW as a proxy for emission line strength. It is indeed a good approximation as it is one of the strongest emission lines and directly relates to our SF project. However, the shortcoming of this method is that it relies on getting redshift correct ...

Furthermore, for very low SNR data where continuum cannot be measured, it is difficult to robustly measure EW. However, similarly relies on getting redshift correct.

Having a metric that characterizes strength of emission lines (similar to what we were trying to get with H α EW) that does not rely on redshift would be good. Something like number of "peaks" in observed spectra could work? Still thinking on this front ...

Use random forest to test what are best proxies - Zhiwei Shao

Q: How shall we use purity and completeness weights in our analysis?

A: Even in the scenario where we define some success rate by a DChi2 cut and use that to weight the successful redshifts in downstream science analysis, we will need to understand the purity and completeness weights. Thus, we forego defining a redshift success rate and instead use the entire dwarf galaxy spectroscopic catalog* in our analysis. We then weight each point using the purity and completeness fractions.

As we shall show, these fractions are themselves non-trivial functions of fiber magnitude, line EW and Dchi2. w_{spec} is the spectroscopic completeness weight we will apply to each galaxy in dwarf catalog. These weights can be both smaller and larger than 1.

$$w_{\text{spec}} = \frac{w_{\text{pure}}(r_{\text{fib}}, \text{EW}_{H\alpha}, \Delta\chi^2)}{w_{\text{comp.}}(r_{\text{fib}}, \text{EW}_{H\alpha}, \Delta\chi^2)}$$

*this is the catalog of $\text{LogM}_{\text{star}} < 9$ galaxies where the stellar mass is computed assuming z_{RR} is correct

How can we generate realistic theoretical spectra instead of using FSPS?

We train NMF on relatively high SNR, robust redshift spectra. We obtain templates and the distribution of coefficient values. If we have 10 templates, we will have 10 coefficients, and thus will have a 10 dimensional distribution. If we have a reasonable large population of galaxies, then we would sample the whole distribution quite effectively.

We then generate mock spectra by drawing from this distribution, randomizing redshifts, and scale the overall flux to mimic different r_{fiber}-mags. We then mock observe and push through Redrock and NMF to get inferred redshift and NMF.

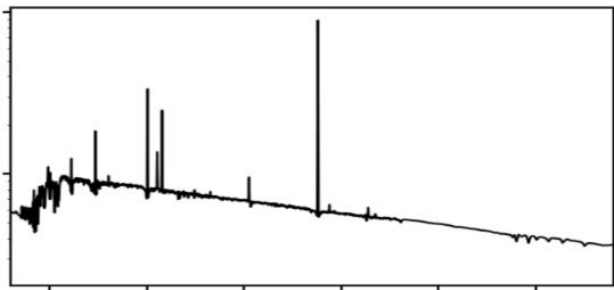
Our simulation based approach for measuring purity and completeness

1. Generate theoretical spectra

Generate a library of theoretical galaxy spectra using FSPS. We sample over a wide range in metallicity, ionization, age (proxy for SFR).

We specifically use the tau SFH parameterization with $\tau=2$ and ages between 5 and 11.5 Gyr. For metallicities, we span -1 to -0.25 and for ionization, we span -3 to -1. We assume a fixed velocity dispersion of 50 km/s and fixed dust law.

The assumption is that these choices do not impact the completeness model, except insofar as they impact H α EW

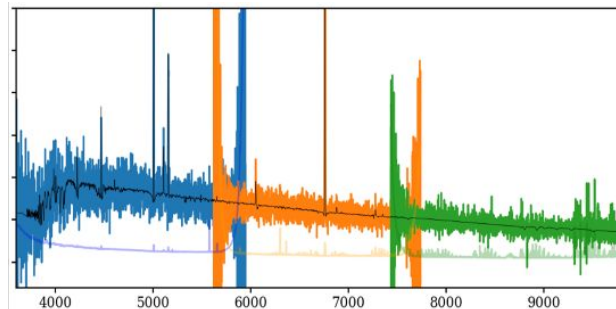


2. Mock observe spectra

Select a random redshift between 0.001 and 0.15 (all of our dwarf sample is in this range), and redshift the spectra.

Convolve spectra with [decamDR1noatm filter](#) and scale overall flux to agree with a random (fiber) magnitude drawn from U[19, 23.5]

Use [feasibgs](#) to mock observe spectra assuming some obs. conditions. For BGS Bright-like, we assume EXPTIME=180s, air_mass = 1, seeing=1.1 with nominal dark time conditions. For LOWZ-Dark, we assume EXPTIME = 900s with rest all the same

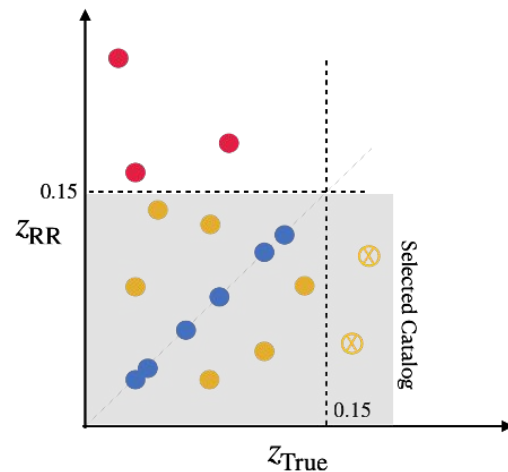


3. Run Redrock

Run Redrock on these mock observed spectra to test recovery of redshift.

Specifically, find under what conditions the redshift is wrong by measuring **purity weights** and **completeness weights**

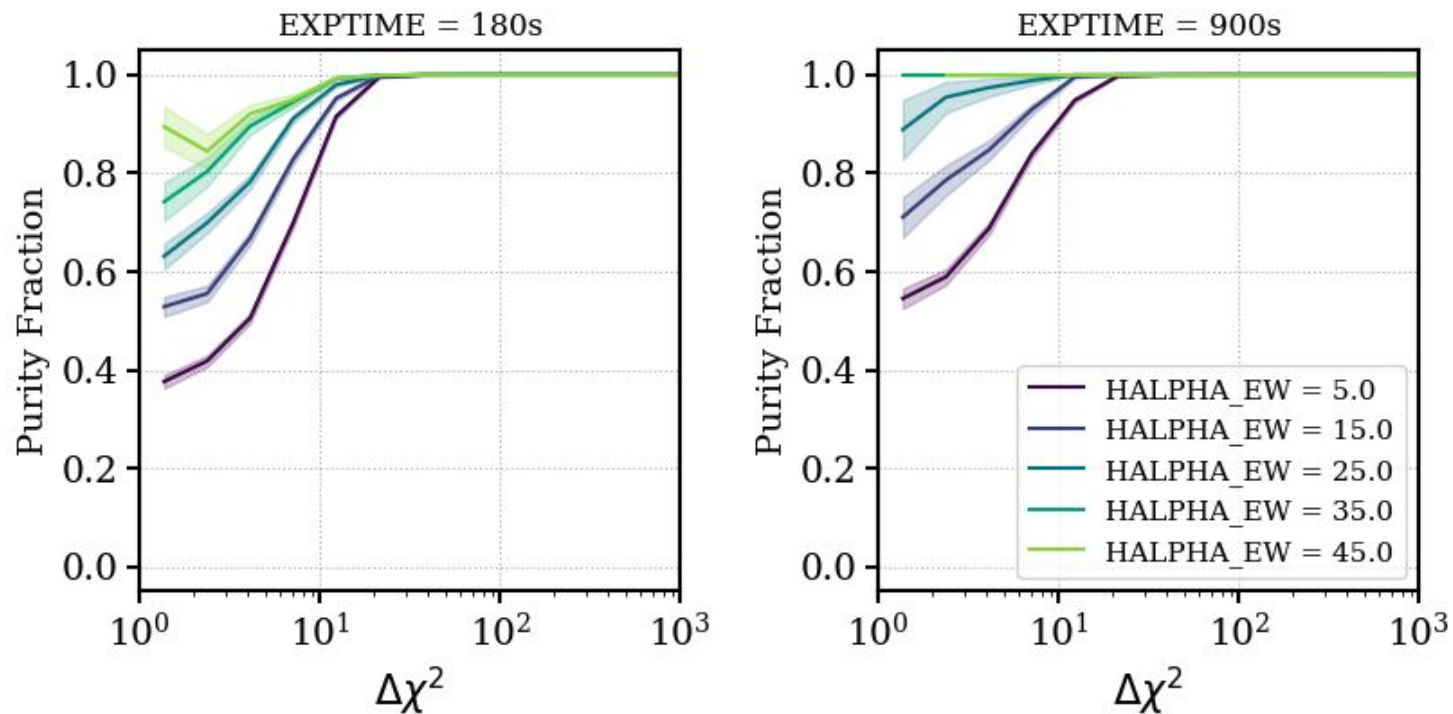
Schematic below of various ways in redshift failures impact our nominal dwarf galaxy sample at $z_{RR} < 0.15$



Make a plot of effective exposure times for BGS Bright and LOWZ Dark

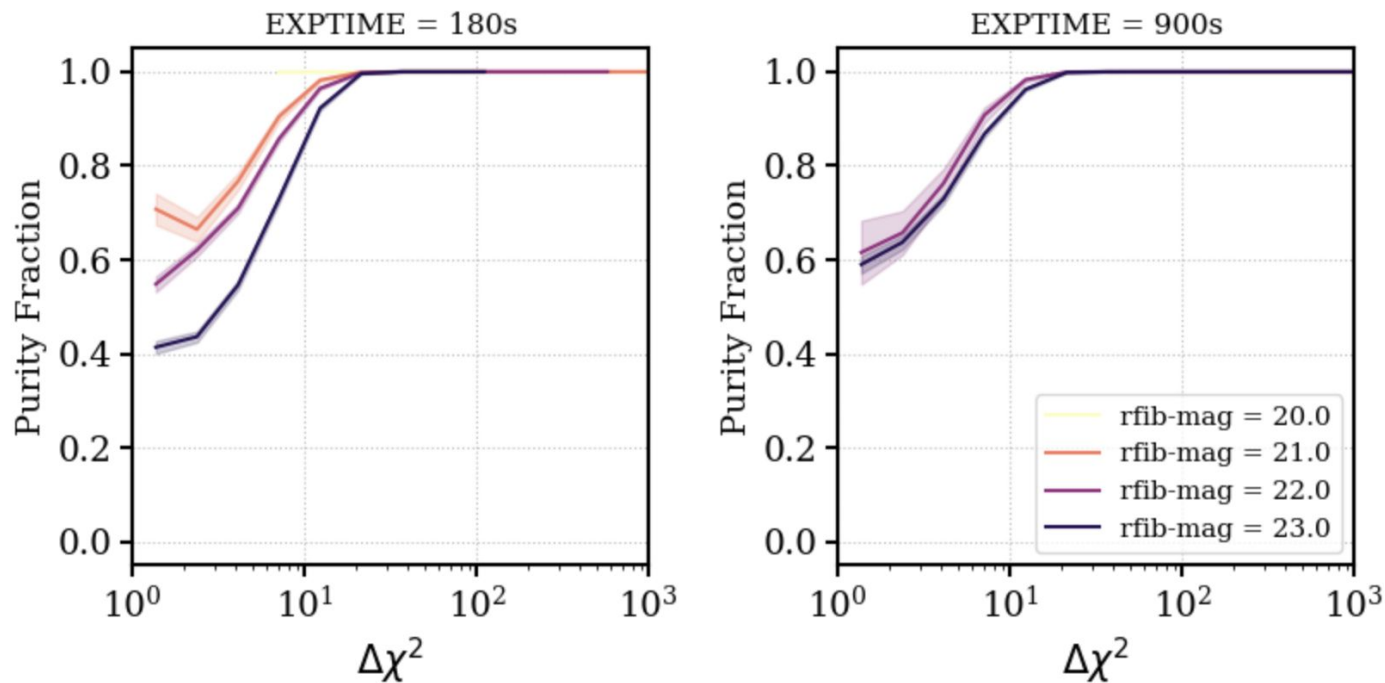
Results from the simulations...

Purity Fractions



Interestingly, it seems that $\Delta\chi^2$ alone does not describe the purity fraction, but depends on EW value (and also on exposure time!). Each line is the purity fraction vs. $\Delta\chi^2$ for bins in Halpha EW (0-10, 10-20 etc.)

Purity Fractions

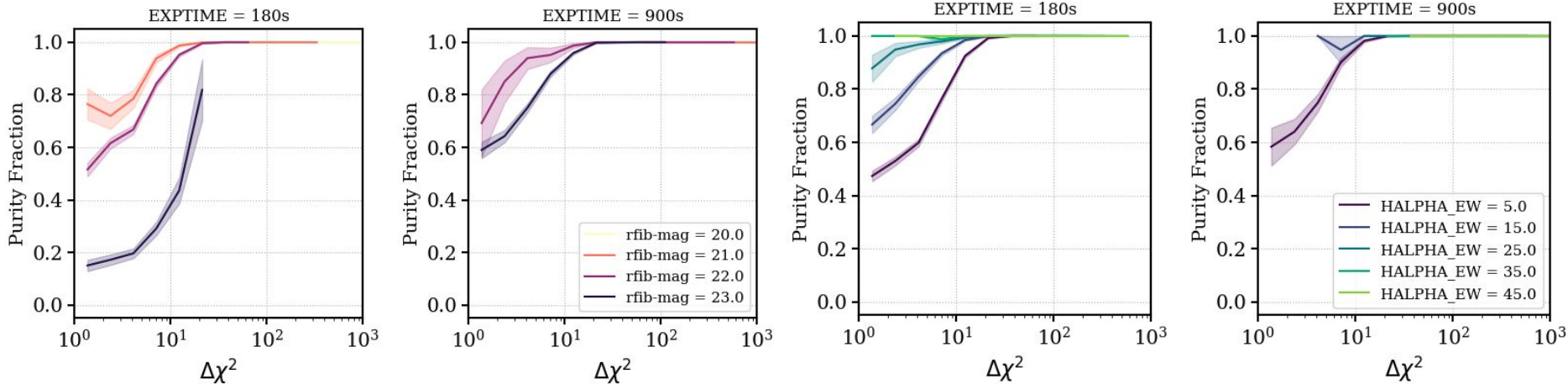


Similarly for fiber-magnitude. Each line is the purity fraction vs. $\Delta\chi^2$ for bins in fiber mag (19.5 - 20.5 etc.)

Purity Fractions

$5 < \text{EW} < 10$

$21.5 < \text{fib} < 22.5$



Note that fiber magnitude and Halpha EW are not capturing the same trends. The above plot shows that when conditioning on one parameter, the purity fraction still depends significantly on the other parameter.

One caveat in purity fractions modeling...

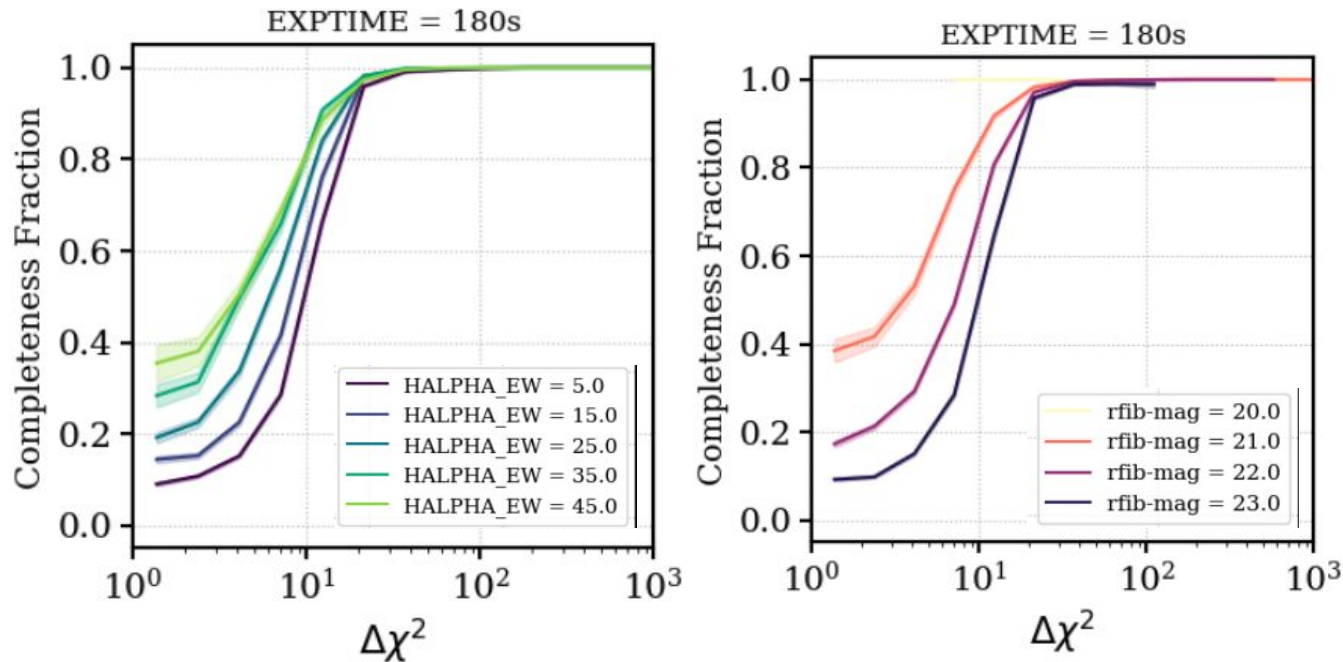
Note that in computing purity fractions here, I have not considered the population of higher redshift objects with incorrect lower redshifts ($z_{RR} < 0.15$ & $z_{True} > 0.15$). I have not considered this component in my simulations. This quantity is in the purity fraction definition (see slide 8)

However, looking at SV3 data, it seems to be reasonable to not account for this as the effect is very small. For example, there are 17 galaxies with incorrect redshifts with $z_{RR} < 0.15$ & $z_{True} < 0.15$, but only 1 galaxy with incorrect redshift with $z_{RR} < 0.15$ & $z_{True} > 0.15$

So it seems okay to not account for this in our simulations ...

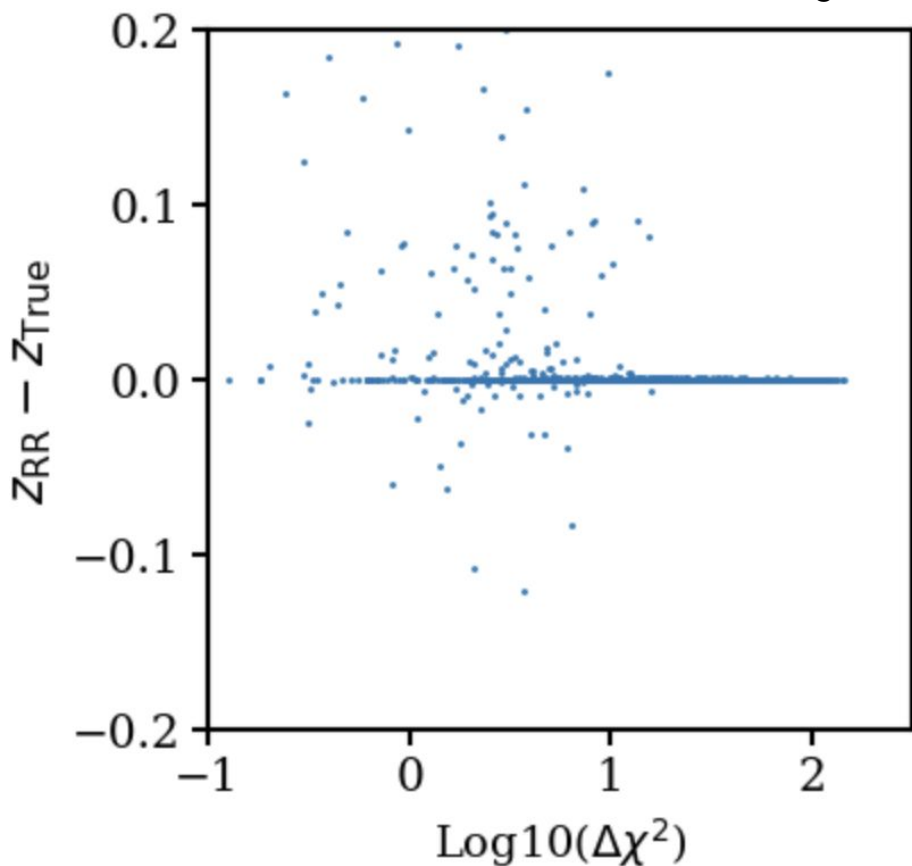
Completeness Fractions

The story is very similar with completeness fractions ...



Note how the values of completeness fractions are lower than purity fractions. This is because in completeness fraction we are considering incorrect redshifts ($z_{RR} > 0.15$ & $z_{True} < 0.15$) that were not being considered in purity fractions calculations ($z_{RR} < 0.15$ & $z_{True} < 0.15$).

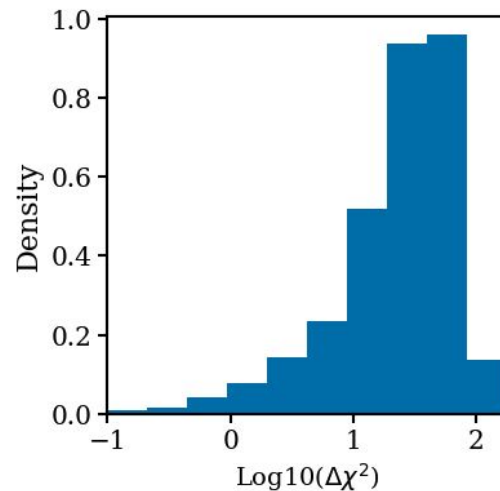
EXPTIME=180s, $5 < \text{EW} < 7.5$ & $21 < \text{r-fib-mag} < 21.5$.



We still need to include Dchi2 in model as EW + fiber mag does not describe the full behavior!!

When restricting to a specific bin in fiber mag and EW, there is still variation in DChi2, and this scatter is relevant to redshift robustness.

Example on the left is for simulated exptime=180s spectra for $5 < \text{EW} < 7.5$ and $21 < \text{r-fib-mag} < 21.5$.



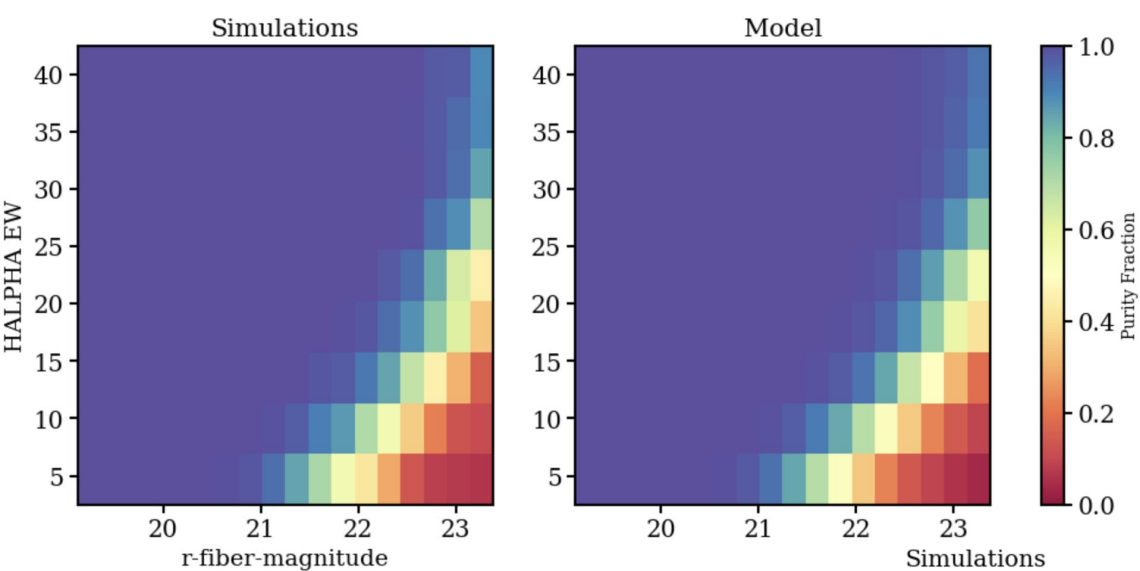
So our model for purity fraction and completeness fraction depends on 3 variables and also further influenced by exposure time (180s vs. 900s). It would be non-trivial (and too tedious) to obtain 3D analytic expressions for these by using sigmoids etc. Thus, we shall use a neural network!

Inspiration for this is the paper [Photometric Completeness Modelled with Neural Networks](#) by William Harris and Joshua Speagle.

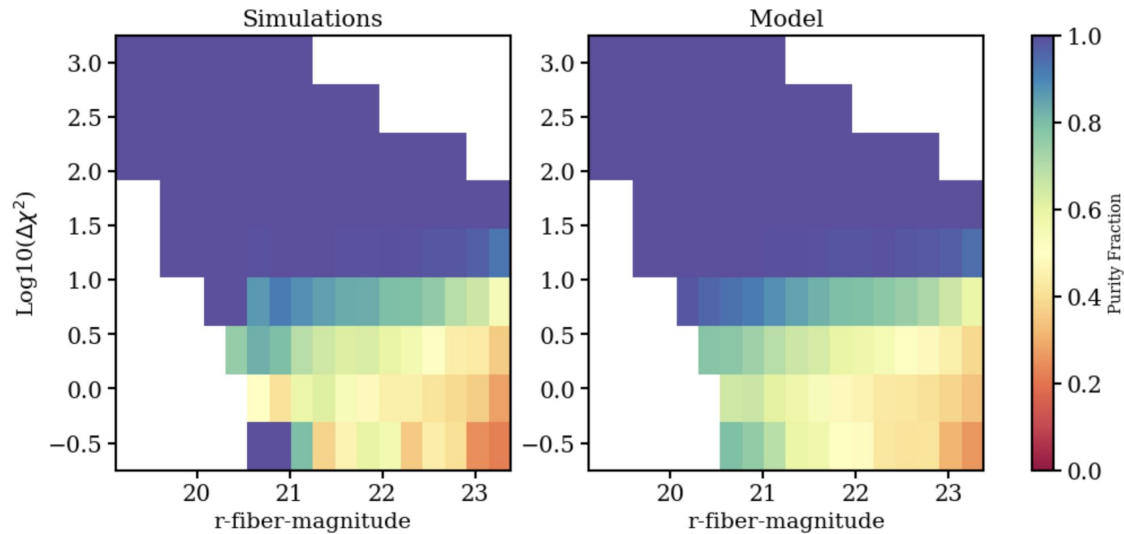
They model the photometric completeness of stars in HST images as a function magnitude, color, sky noise and crowding and find that a NN is able to accurately model the completeness function.

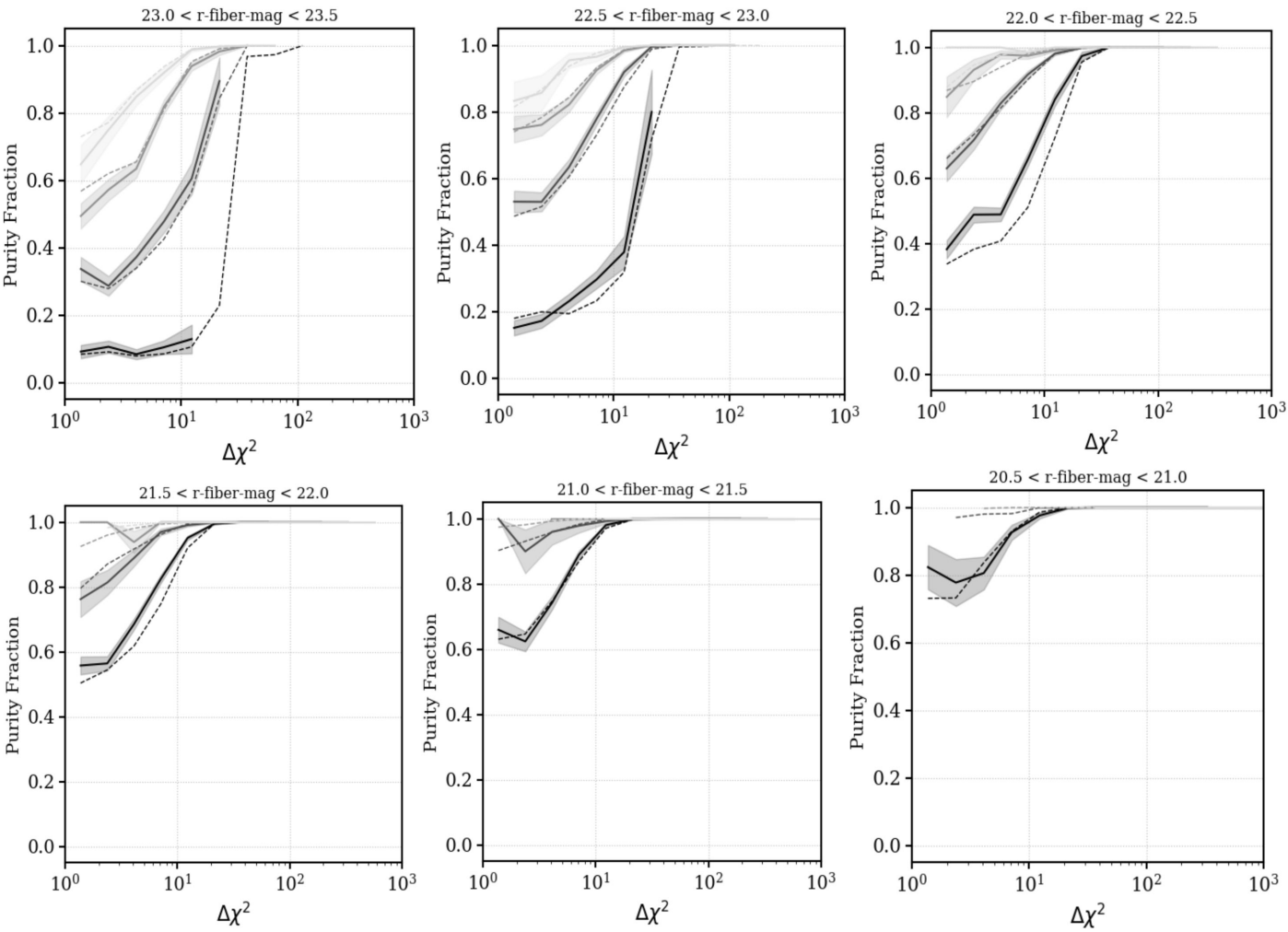
Their task is very similar to the one we have at hand. In their paper, the training set of $\sim 260k$ simulations of stellar recovery (binary "0" or "1") for different parameters (described above).

For us, the binary variable will be whether redshift is correct or not. For purity fraction, we will restrict training set to $z_{RR} < 0.15$, while for completeness fraction, we will use our full simulation set ($z_{True} < 0.15$)



After training the model, some validation plots ...





The different lines are different bins of Halpha EW. Solid line is simulation and dashed line is model

So this approach seems to work well. Once other wrinkles have been ironed out, I can run a larger set of spectral simulations and retrain the model.

One caveat when applying to real data ...

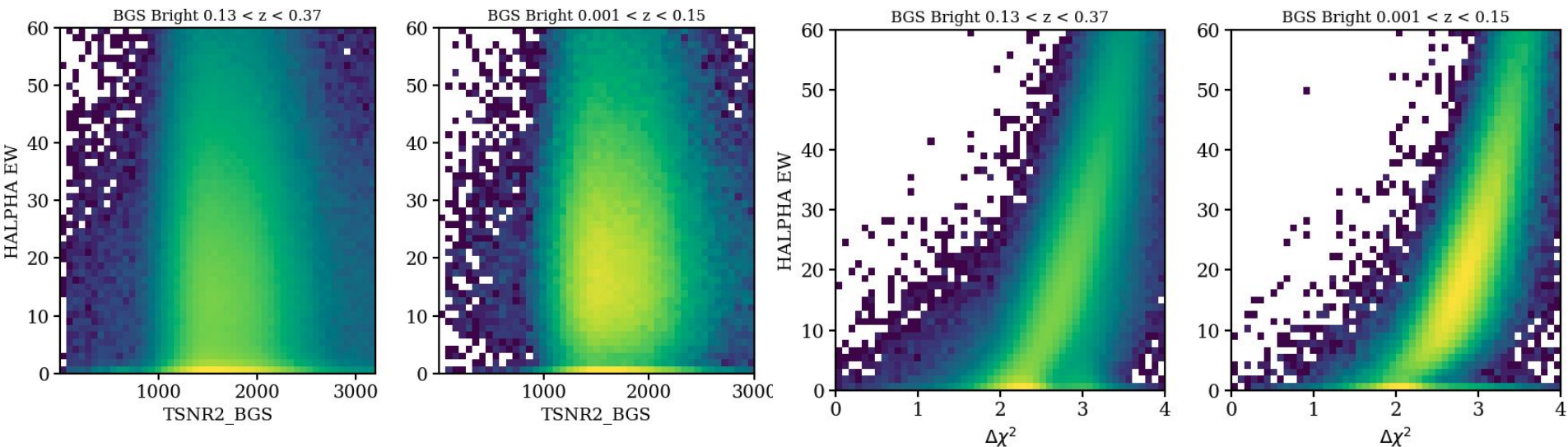
This model is being constructed on the true H α EW, however, in reality when redshift goes wrong, we will also be measuring the incorrect EW.

I could include this as part of the model, by using "observed EW" using z_{RR} instead of true EW.

This is a complication that I will deal with later as the general framework does not change significantly.

Appendix A: TSNR2 correlations with HALPHA EW

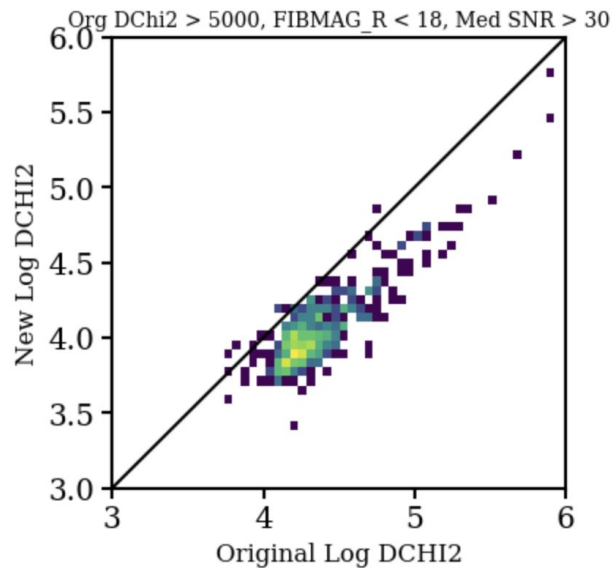
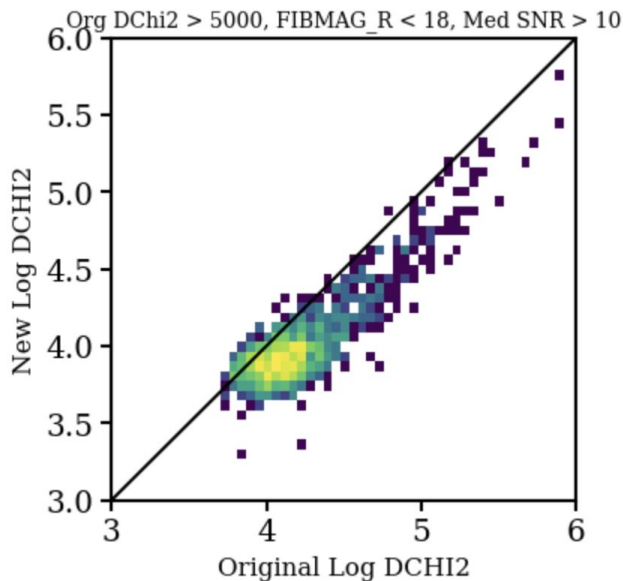
These are BGS Bright galaxies in those specific redshift ranges with $20.5 < r_{\text{fib}} < 21$. Dchi2 correlates much stronger with HALPHA EW unlike TSNR2 as one might expect ...



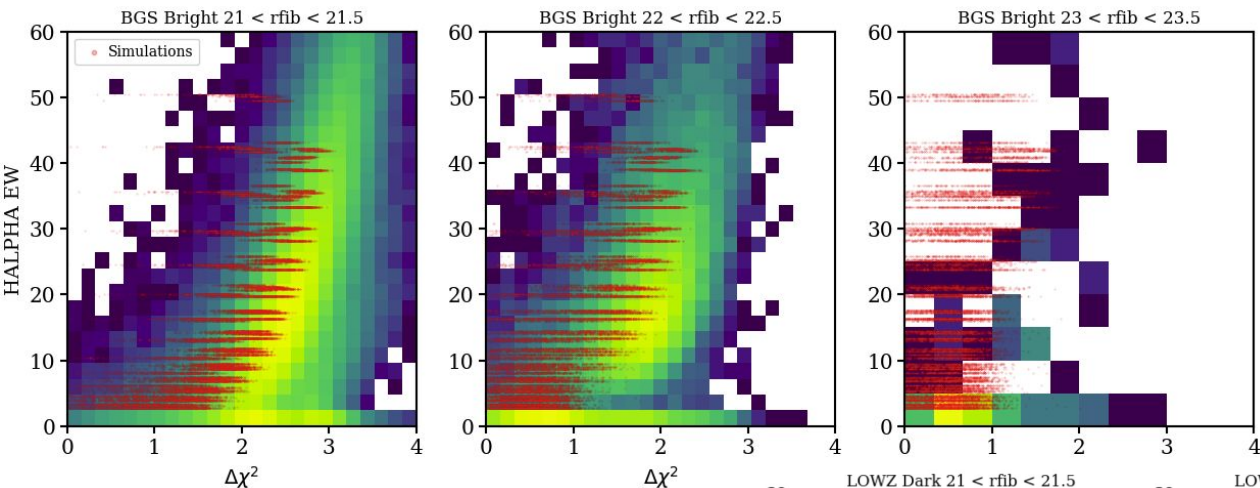
Q: Are real BGS Bright observations similar to these conditions?

To answer the above, I can try 2 things: 1) I take very high SNR DESI BGS Bright spectra (r_{fib} < 18, D_{chi2} > 5000), and reobserve them using my setup + run redrock on them to see how new D_{chi2} values compare to original. 2) See how D_{chi2} vs. EW plot look in simulations vs. real data.

- 1) The fact that we are consistently underpredicting the D_{chi2} value in our simulations could be due to the fact that we are just adding noise to the data by simulating obs. again, or exposure time is smaller than reality ...

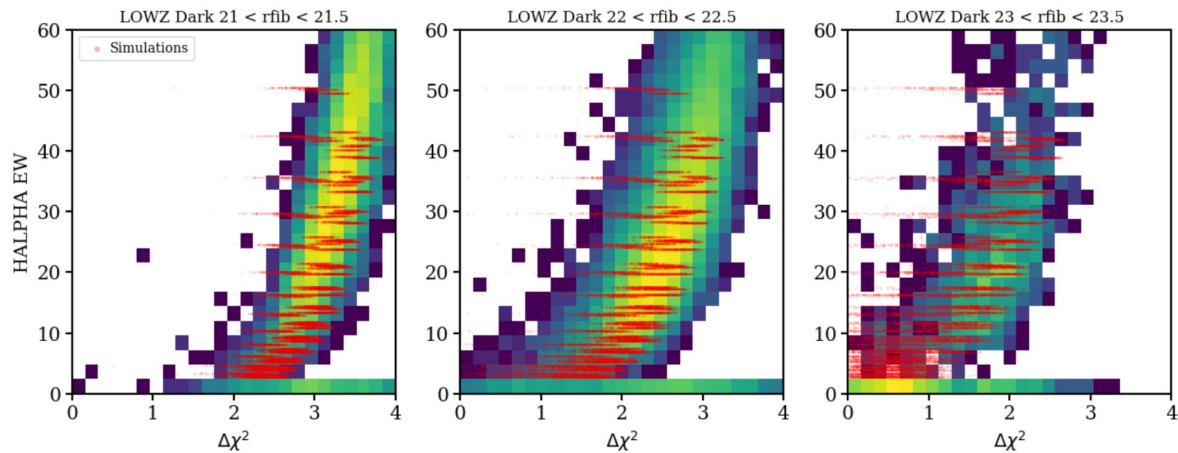


2) Here we compare the Dchi2 - EW plot for real data and our spectral simulations for both BGS Bright and LOWZ Dark like observing conditions



Red points are our spectral sims.

The larger spread in BGS is due to a larger spread in effective exposure time



2) It seems that for LOWZ Dark-like conditions, our simulations line up pretty well with real observations. However, for BGS Bright-like conditions, our spectral simulations under predict the value of DChi2 at a given EW. With a slightly longer exposure time (maybe ~200s-250s?), they could be in better agreement.

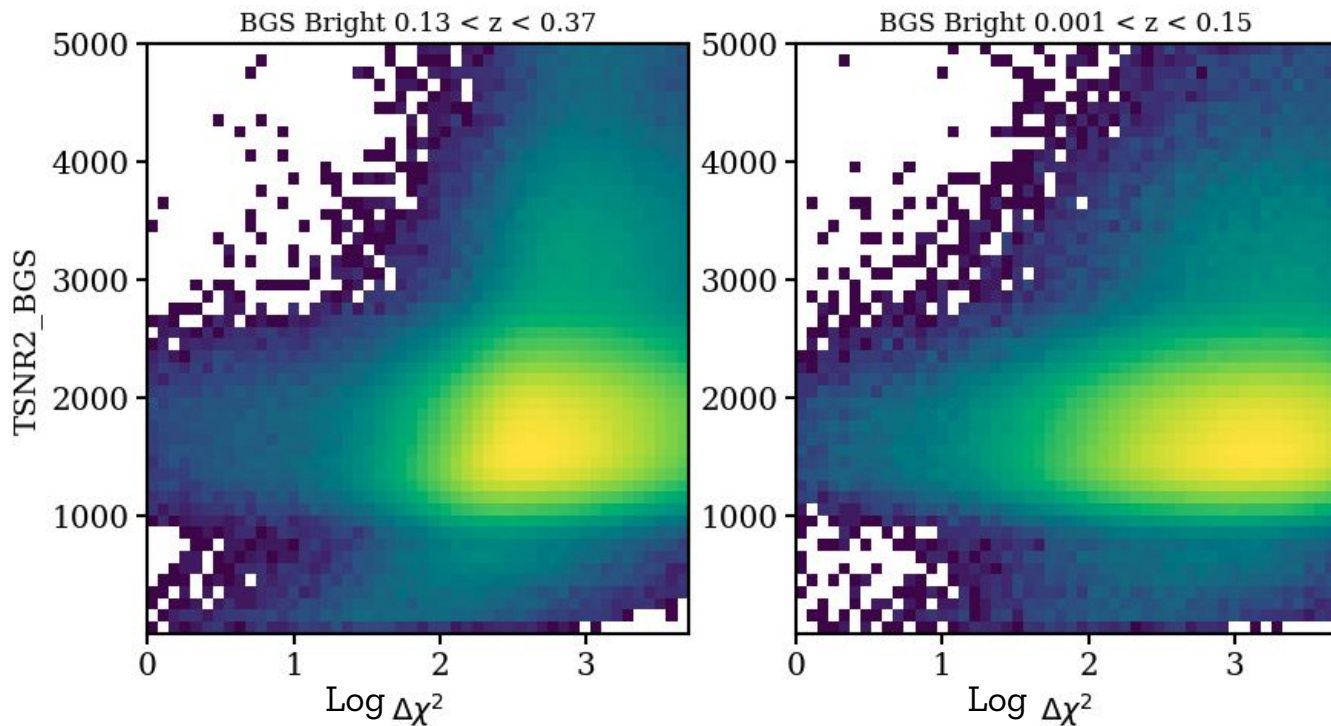
Another interesting observation is the decreasing DChi2 at $EW > \sim 40-45$. One would not expect this to happen as stronger EW would mean more confident redshift. However, in my spectral simulations, spectra with very strong EW (small age in fsps) tend to have very blue slopes at $< \sim 4000\text{\AA}$ and are not fit well by Redrock, hence the decreasing DChi2 (and increasing Chi2)

Some notes from discussion with Julien Guy:

- 1) Compare EXPTIME with the effective exposure (BGS eff time) of the surveys (this is similar to looking at zenith + 1" seeing), for dark time we can really well to 1000s Effective time within 10%. Hard to do for Bright time to maintain a constant effective time for BGS, look at the effective time in the data)
- 2) Feasibgs noise model is too optimistic, does not include noise correlations, residual cosmic rays, bias level, paint theory spectra + sky fiber spectra (similar TSNR2 values of my galaxies) + galaxy poisson noise etc., will need desi-sim, this was very important for LBG galaxies that are very faint ($r \sim 24$) but might not be very relevant for our bright galaxies)

Does TSNR2 metric correlate with Dchi2 (and thus redshift success rate) ?

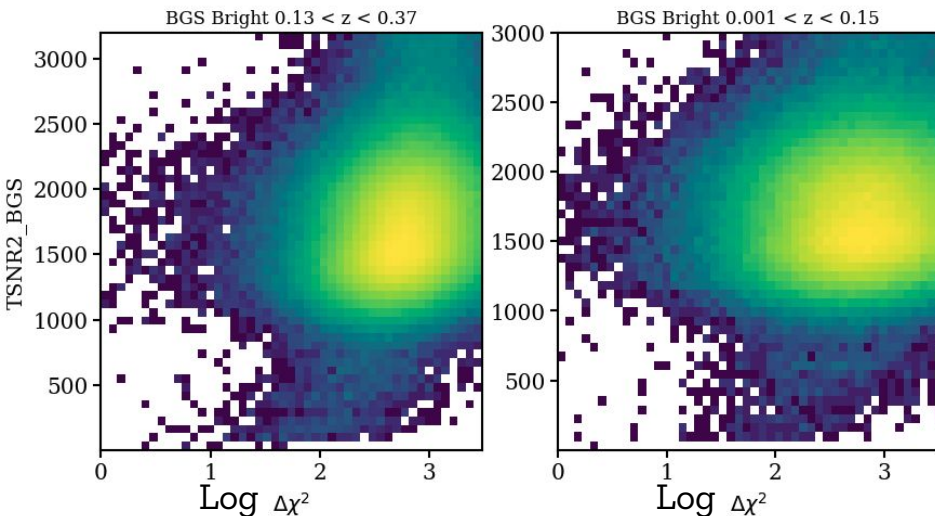
These are BGS Bright galaxies in those specific redshift ranges and no additional cuts



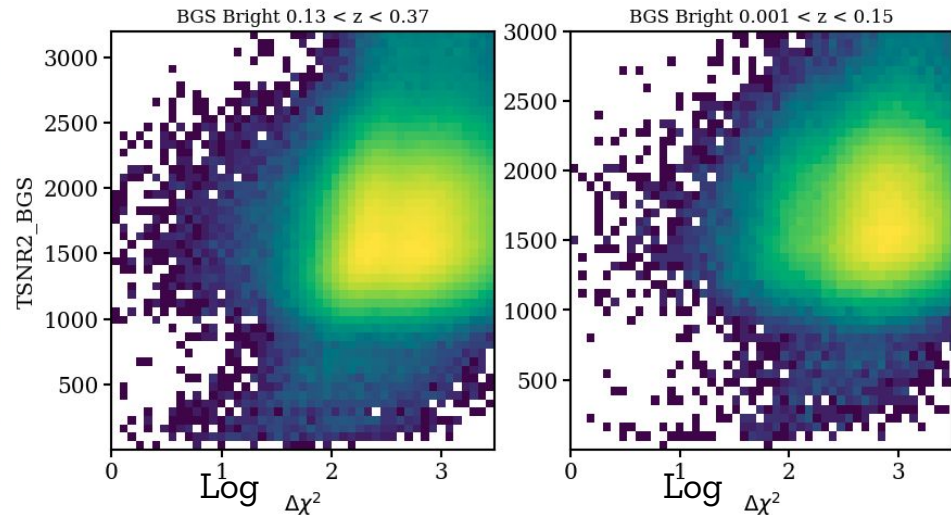
Does TSNR2 metric correlate with Dchi2 (and thus redshift success rate) ?

These are BGS Bright galaxies in those specific redshift ranges conditioned on some variables

with $10 < \text{HALPHA_EW} < 20$

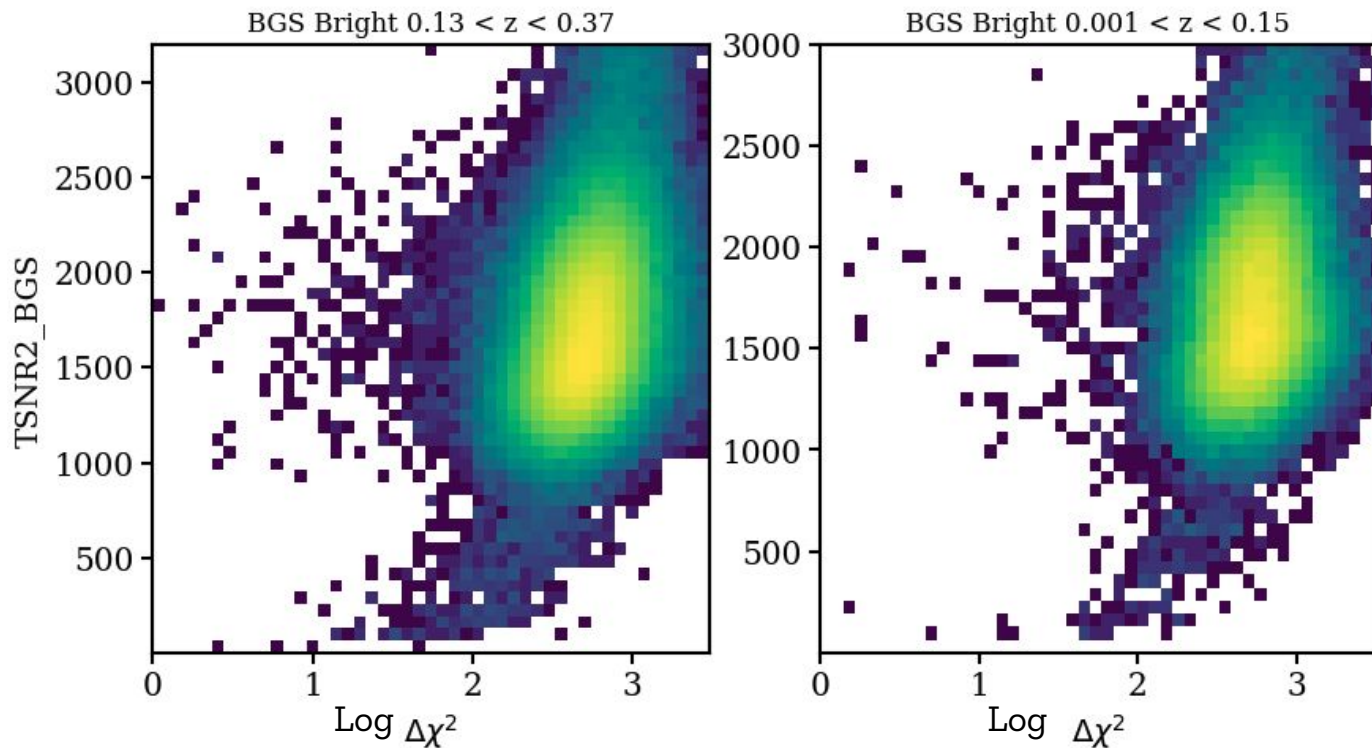


with $20.5 < \text{fiber mag} < 21$



Does TSNR2 metric correlate with Dchi2 (and thus redshift success rate) ?

These are BGS Bright galaxies in those specific redshift ranges with $20.5 < r_{\text{fib}} < 21$ AND $10 < \text{HALPHA EW} < 20$



Does TSNR2 metric correlate with Dchi2 (and thus redshift success rate) for low-redshift BGS?

These plots suggest a few things:

- i) Unlike the LRGs, it seems that there is no strong correlation between TSNR and Dchi2 for BGS (only seen with additional conditioning); see Figure 44 in [Guy et al 2023](#).
- ii) When conditioning on fiber mag and Halpha EW, TSNR2 and Dchi2 are roughly equivalent quantities.
- iii) Separately (see Appendix A at end of slide deck), I plot TSNR2 vs. HALPHA EW, conditioned on fiber mag and find no strong correlation. I would have expected this given that TSNR2 should be characterizing strength of spectral features? Dchi2 correlates much stronger with HALPHA_EW.

Given these observations, and the fact that we do expect DChi2 to be a direct indicator of redshift robustness, we choose to model our spectroscopic completeness as a function of Dchi2, HALPHA EW and fiber-mag. It seems that we could also replace TSNR2 with Dchi2 here, but for sake of simplicity, we choose to use Dchi2.